

Test of Mathematics for University Admission
Mock Set I Paper 2
 Mathematical Reasoning

Name: _____

Date sat: _____

Score: _____ / 20

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Throughout, “must be true” means true in every case allowed by the information given, and a counterexample must satisfy the hypothesis while failing the conclusion.

[illegible]

Question 1

Two fair six-sided dice are rolled. Consider the events A : the first die shows an even number; B : the total is 7; C : the total is 6; D : both dice show even numbers. Which of the following pairs of events are independent?

- I A and B
 - II A and C
 - III A and D
 - IV B and D
- A I only
B I and II only
C I and III only
D I and IV only
E III only
F none of them

Question 2

A function f , defined for all $x > 0$, is said to *tend to infinity* if for every number M there is a number N such that $f(x) > M$ for every $x > N$. How many of the following functions tend to infinity?

- I $f(x) = x - \sin x$
 - II $f(x) = x \sin x$
 - III $f(x) = (\ln x)^2$
 - IV $f(x) = xe^{-x}$
 - V $f(x) = x + 10 \sin x$
- A 1
B 2
C 3
D 4
E 5

Question 3

A polynomial f satisfies $f(x) = f(2 - x)$ for every real x , and $\int_0^2 f(x) dx = 6$. Which of the following are determined by this information alone, with the value shown?

- I $\int_0^2 xf(x) dx = 6$
 - II $\int_0^2 (x - 1)^3 f(x) dx = 0$
 - III $\int_0^2 x^2 f(x) dx = 8$
 - IV $f'(1) = 0$
- A I only
B I and II only
C I, II and III only
D II and IV only
E all four
F I, II and IV only

Question 4

A student proves that $|a + b| \leq |a| + |b|$ for all real a and b , using cases.

- (I) If $a \geq 0$ and $b \geq 0$, then $|a + b| = a + b = |a| + |b|$.
- (II) If $a < 0$ and $b < 0$, then $|a + b| = -(a + b) = |a| + |b|$.
- (III) If $a \geq 0 > b$, then $a + b < a \leq |a| + |b|$, so $|a + b| \leq |a| + |b|$.
- (IV) The case $b \geq 0 > a$ is the same as (III) with a and b swapped.

Which of the following is correct?

- A** The proof is complete and correct.
- B** The claim is false when a and b have opposite signs.
- C** Case (II) is wrong, because $|a + b| = a + b$ there.
- D** The claim is true, but case (III) only shows $a + b \leq |a| + |b|$; it does not rule out $-(a + b) > |a| + |b|$.
- E** Case (IV) is invalid because the claim is not symmetric in a and b .
- F** The cases do not cover every pair (a, b) .

Question 5

The sequence $u_1, u_2, \dots$ is defined by $u_1 = 1$ and

$$u_{n+1} = \frac{1}{2} \left(u_n + \frac{4}{u_n} \right).$$

Which of the following are true?

- I** $u_n > 2$ for every $n \geq 2$.
 - II** $u_{n+1} < u_n$ for every $n \geq 2$.
 - III** Every term is rational.
 - IV** $u_5 = 2$
- A** I and II only
 - B** I, II and III only
 - C** III only
 - D** I, II, III and IV
 - E** I and III only
 - F** II and III only

Question 6

For the cubic $p(x) = x^3 + ax^2 + bx + c$ with real coefficients, which of the following conditions guarantee that p has three real roots (counted with multiplicity)?

- I** $p(-1) > 0$ and $p(1) < 0$
 - II** $a^2 < 3b$
 - III** $b < 0$ and $c = 0$
 - IV** $a = 0$ and $b = 0$
- A** I only
 - B** I, II and III only
 - C** III only
 - D** I and IV only
 - E** I and III only
 - F** II and IV only

Question 7

For which of the following equations is it true that there is exactly one real solution?

- I $e^x = 1 + x$
- II $e^x = 2x$
- III $\ln x = x - 1$
- IV $2^x = x^2$
- A I only
- B I, II and III only
- C III only
- D I, III and IV only
- E I and III only
- F II and IV only

Question 8

A bag contains r red balls and b blue balls, where $r, b \geq 1$. Two balls are drawn at random without replacement. Which of the following are true?

- I The two balls are equally likely to be the same colour as different colours exactly when $(r-b)^2 = r + b$.
- II If $r = b$, the probability that the two balls are different colours is greater than $\frac{1}{2}$.
- III It is possible for the probability that both balls are red to equal exactly $\frac{1}{2}$.
- IV The probability that the two balls are different colours never exceeds $\frac{2}{3}$.
- A I only
- B I and II only
- C II and III only
- D I, II and III only
- E all four
- F I, III and IV only

Question 9

Which of the following counts are correct?

- I There are 4 strictly increasing functions from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$.
- II There are 20 non-decreasing functions from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$.
- III There are 36 functions from $\{1, 2, 3, 4\}$ onto $\{1, 2, 3\}$ (every value is taken).
- IV There are 12 one-to-one functions from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$.
- A I, II and III only
- B I and II only
- C I and IV only
- D all four
- E II and III only
- F I only

Question 10

A triangle has sides $a \leq b \leq c$ that form an arithmetic progression, so $a + c = 2b$. Let B and C be the angles opposite b and c . Which of the following must be true?

- I $B \leq 60^\circ$
- II The triangle is not right-angled.
- III $3a > c$
- IV $C < 90^\circ$
- A I only
- B I, III and IV only
- C I and III only
- D III only
- E I, II and III only
- F II and IV only

Question 11

A bag contains n balls numbered $1, 2, \dots, n$, where $n \geq 2$. Two balls are drawn at random without replacement. For which values of n is the probability that the sum of the two numbers is even exactly $\frac{7}{15}$?

- A $n = 15$ only
- B $n = 16$ only
- C $n = 14$ and $n = 15$
- D $n = 30$ only
- E $n = 15$ and $n = 16$
- F no value of n

Question 12

Without evaluating them exactly, decide which of the following integrals are strictly positive.

- I $\int_0^\pi x \sin x \, dx$
- II $\int_0^{2\pi} x \sin x \, dx$
- III $\int_{-1}^1 x^3 e^{x^2} \, dx$
- IV $\int_0^{2\pi} \frac{\sin x}{1+x} \, dx$

- A I only
- B I and II only
- C I, II and IV only
- D I and IV only
- E IV only
- F I, III and IV only

Question 13

A function f is differentiable on $[0, 1]$, with $f(0) = 0$, $f(1) = 1$ and $f'(x) \leq 1$ for every x in $[0, 1]$. Which of the following must be true?

- I $f(x) = x$ for every x in $[0, 1]$.
- II $f(2) \leq 2$
- III $f'(\frac{1}{2}) = 1$
- IV $\int_0^1 f(x) dx = \frac{1}{2}$

- A I only
- B I, III and IV only
- C I and IV only
- D all four
- E III and IV only
- F I, II and IV only

Question 14

A continuous function f on $[0, 2]$ has $F(t) = \int_0^t f(x) dx$. It is known that $F(t) \geq 0$ for every t in $[0, 2]$ and $F(2) = 0$. Which of the following must be true?

- I $f(0) \geq 0$
- II $f(2) \leq 0$
- III $f(c) = 0$ for some c in $[0, 2]$.
- IV $f(1) = 0$

- A III only
- B I and II only
- C I and III only
- D all four
- E II and III only
- F I, II and III only

Question 15

Integers $x < y < z$ satisfy $x + y + z = xyz$. Which of the following must be true?

- I There are infinitely many such triples.
- II If x, y and z are all positive, then $(x, y, z) = (1, 2, 3)$.
- III $xyz \geq 0$
- IV If none of x, y, z is zero, then $|x|, |y|, |z|$ are 1, 2, 3 in some order.

- A I only
- B II and IV only
- C I, II and III only
- D I, II and IV only
- E I and III only
- F all four

Question 16

Let $f(x) = x^4 + ax^3 + bx^2$, where a and b are real. Which of the following conditions guarantee that f has exactly one stationary point?

- I $a = 0$ and $b > 0$
- II $9a^2 < 32b$
- III $b > 0$
- IV $a = 0$ and $b = 0$
- A I only
- B I and II only
- C I, II and IV only
- D I, II, III and IV
- E II only
- F III and IV only

Question 17

A function f is twice differentiable, with $f''(x) > 0$ for every real x and $f(0) = 0$. Which of the following must be true?

- I $f(2) > 2f(1)$
- II $f(1) + f(-1) > 0$
- III $f(x) \geq f'(0)x$ for every real x .
- IV $f'(x) > 0$ for every $x > 0$.
- A I only
- B I, II and III only
- C I and III only
- D I, II, III and IV
- E II and III only
- F III and IV only

Question 18

For real constants p and q , consider the equation

$$x|x| + px + q = 0.$$

Which of the following statements are true?

- I If $p \geq 0$, the equation has exactly one real solution.
- II If $p < 0$ and $q = 0$, the equation has exactly three real solutions.
- III For some values of p and q the equation has exactly two real solutions.
- IV For some values of p and q the equation has exactly four real solutions.
- A I, II and III only
- B I and II only
- C II and III only
- D all four
- E I only
- F I, III and IV only

Question 19

Here *increasing* means that $u < v$ implies $f(u) < f(v)$. Which of the following statements are true?

- I If f is differentiable everywhere and $f'(x) > 0$ for every $x \neq 0$, then f is increasing.
- II If f is continuous everywhere, differentiable except possibly at 0, and $f'(x) > 0$ for every $x \neq 0$, then f is increasing.
- III If f is differentiable except at 0, and $f'(x) > 0$ for every $x \neq 0$, then f is increasing.
- IV If f is differentiable and increasing, then $f'(x) > 0$ for every x .

- A I and II only
- B I only
- C I, II and III only
- D II only
- E I, II and IV only
- F all four

Question 20

For real constants a and b , consider the equation

$$\sqrt{x+a} = x+b.$$

Which condition on a and b is necessary and sufficient for the equation to have exactly two distinct real solutions?

- A $a > b - \frac{1}{4}$
- B $b - \frac{1}{4} < a < b$
- C $a \leq b$
- D $b - \frac{1}{4} \leq a \leq b$
- E $b - \frac{1}{4} < a \leq b$
- F $a < b + \frac{1}{4}$

END OF PAPER